

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

.Net Core and Progressive Web Application Practical

Practical – VIII(Curiosity Question)

Roll No.: T016	Name: Himani Malankar
Class:TYIT	Batch:1
Date of Assignment: 10/08/2026	Date/Time of Submission: 10/08/2026

Practical: Student Web API and PWA Integration

1. **What is a Web API?**
2. **What is JSON? Why is JSON commonly used for transferring data through Web APIs?**
3. **What is the purpose of the HTTP GET method in a Web API?**
4. **Create a *Student* model with the following properties:**
 - Roll No
 - Student Name
 - Course
5. **Write an ASP.NET Core Web API that returns the following student data in JSON format:**
 - Roll No: **101**
 - Name: **Amit Sharma**
 - Course: **B.Sc. IT**
6. **Write an HTML page that contains a link to *manifest.json*.**
7. **Write JavaScript code using *fetch()* to call the Student Web API and display the student details on the HTML page.**

CODE:

(*Student.cs*):

```
namespace Practical_8_CQm.Models
{
    public class Student
    {
        public int RollNo { get; set; }

        public string Name { get; set; }

        public string Course { get; set; }
    }
}
```

(*StudentController.cs*):

```
using Microsoft.AspNetCore.Mvc;

namespace Practical_8A.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    public class StudentController : ControllerBase
    {
    }
```

```
[HttpGet]
public IActionResult Get()
{
    return Ok(new
    {
        RollNo = 101,
        Name = "Amit Sharma",
        Course = "B.Sc. IT"
    });
}
```

(Index.cshtml):

```
<h2>Student Record</h2>

<button type="button" onclick="loadStudent()">Load Student</button>

<p id="data"></p>

<script>function loadStudent() {

    fetch("/api/student")
    .then(response => response.json())
    .then(student => {

        document.getElementById("data").innerHTML =
            "Roll No : " + student.rollNo +
            "<br>Name : " + student.name +
            "<br>Course : " + student.course;

    });

}</script>
```

(manifest.json):

```
{
  "name": "Student PWA (T016)",
  "short_name": "Student",
  "start_url": "/",
  "display": "standalone"
}
```

(_Layout.cshtml):

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>@ViewData["Title"] - Practical 8_CQ</title>
  <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
  <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
  <link rel="stylesheet" href="~/Practical 8_CQ.styles.css" asp-append-version="true" />
  <link rel="manifest" href="/manifest.json" />
</head>
<body>
  <header>
    <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white
border-bottom box-shadow mb-3">
      <div class="container-fluid">
        <a class="navbar-brand" asp-area="" asp-controller="Home"
asp-action="Index">Practical 8_CQ</a>
        <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target=".navbar-collapse" aria-controls="navbarSupportedContent"
aria-expanded="false" aria-label="Toggle navigation">
          <span class="navbar-toggler-icon"></span>
        </button>
      </div>
    </nav>
  </header>
```

```

<div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
  <ul class="navbar-nav flex-grow-1">
    <li class="nav-item">
      <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Index">Home</a>
    </li>
    <li class="nav-item">
      <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
    </li>
  </ul>
</div>
</div>
</nav>
</header>
<div class="container">
  <main role="main" class="pb-3">
    @RenderBody()
  </main>
</div>

<footer class="border-top footer text-muted">
  <div class="container">
    &copy; 2026 - Practical_8_CQ - <a asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
  </div>
</footer>
<script src="~/lib/jquery/dist/jquery.min.js"></script>
<script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
<script src="~/js/site.js" asp-append-version="true"></script>
@await RenderSectionAsync("Scripts", required: false)
</body>
</html>

```

OUTPUT:Pretty print ☐

```
{"rollNo":101,"name":"Amit Sharma","course":"B.Sc. IT"}
```

